Charged Scalar Matter Coupled to Light: Superradiance, Holographic Storage and Driven Density Patterns, as Theorems

The holographic brain paper, rewritten as mathematics

2026-11-05

Source. Nishiyama, Tanaka, Tuszynski, Tsenkova, *Holographic Brain Theory: Super-Radiance, Memory Capacity and Control Theory*, Int. J. Mol. Sci. **25**, 2399 (2024). Equation labels of the form (An) refer to its appendices; labels of the form ($k.n$) are ours.

What this is. The same model, the same four calculations, written so that every assumption is a numbered hypothesis, every result is a proposition with a proof, every misprint is corrected, and every number is derived. Nothing is claimed that is not proved. Section 6 states, as mathematics, what the model does not contain.

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0. Conventions

Units $\hbar = c = 1$. Coordinates $x = (x^0, \mathbf{x})$, $\partial_0 = \partial/\partial x^0$, $\nabla = (\partial_1, \partial_2, \partial_3)$. All fields are smooth and either decay at spatial infinity or are periodic on a box $V = L_1 L_2 L_3$; all integrals are over that box.

Fields: a complex scalar φ; a scalar potential A^0; a vector potential $\mathbf{A} = (A_1, A_2, A_3)$, the covariant spatial components as in the source. The electric field is

$$\mathbf{E} \equiv \partial_0 \mathbf{A} - \nabla A^0,$$

and $-\tfrac{1}{4} F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}(\mathbf{E}^2 - (\nabla \times \mathbf{A})^2)$.

Polar form: $\varphi = \sqrt{\rho}\, e^{i\beta}$ with $\rho = |\varphi|^2 \geq 0$. The quanta of φ carry charge $-e$. Velocity field:

$$\mathcal{V} \equiv \frac{\nabla\beta - e\mathbf{A}}{m}. \tag{0.1}$$

Constants: $m > 0$ (mass), $e > 0$ (coupling), $U(\mathbf{x})$ a real potential, $\rho_b(\mathbf{x}) \geq 0$ a fixed background charge density. The source takes $\rho_b \equiv 0$; we keep it as a parameter because several results turn on it.

Notation for the plasma frequency: for a mean number density $\bar{n} = N/V$,

$$\omega_p^2 \equiv \frac{e^2 \bar{n}}{m}. \tag{0.2}$$

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1. The model

Definition 1.1 (Lagrangian).

$$\mathcal{L} = \tfrac{1}{2}\left(\mathbf{E}^2 - (\nabla\times\mathbf{A})^2\right) - \rho_b A^0 + \varphi^*\left(i\partial_0 + eA^0\right)\varphi - \frac{1}{2m}\left|(\nabla - ie\mathbf{A})\varphi\right|^2 - U|\varphi|^2. \tag{1.1}$$

With $\rho_b = 0$ this is (A1) up to a total derivative.

Proposition 1.2 (Field equations). The Euler–Lagrange equations of (1.1) are

$$\text{(M)}\quad (i\partial_0 + eA^0)\varphi = -\frac{1}{2m}(\nabla - ie\mathbf{A})^2\varphi + U\varphi,$$

$$\text{(G)}\quad \nabla\cdot\mathbf{E} = \rho_b - e\rho,$$

$$\text{(A)}\quad \partial_0 \mathbf{E} = \nabla^2 \mathbf{A} - \nabla(\nabla\cdot\mathbf{A}) + e\rho\,\mathcal{V}.$$

Every solution of (M) satisfies the continuity equation

$$\text{(C)}\quad \partial_0 \rho = -\nabla\cdot(\rho\,\mathcal{V}).$$

Proof. Variation in φ^* gives (M). Variation in A^0: $\partial\mathcal{L}/\partial A^0 = -\rho_b + e\rho$ and $\partial\mathcal{L}/\partial(\partial_i A^0) = -E_i$, so $\nabla\cdot\mathbf{E} = \rho_b - e\rho$. Variation in A_i: $\partial\mathcal{L}/\partial A_i = \frac{e}{m}\rho(\partial_i\beta - eA_i) = e\rho\mathcal{V}_i$ (expand $|(\nabla - ie\mathbf{A})\varphi|^2$ and use $\operatorname{Im}(\varphi^*\partial_i\varphi) = \rho\,\partial_i\beta$), $\partial\mathcal{L}/\partial(\partial_0 A_i) = E_i$, $\partial\mathcal{L}/\partial(\partial_j A_i) = -(\partial_j A_i - \partial_i A_j)$; the Euler–Lagrange equation is $\partial_0 E_i - \partial_j(\partial_j A_i - \partial_i A_j) = e\rho\mathcal{V}_i$, which is (A). For (C), multiply (M) by φ^*, subtract the conjugate, and take the imaginary part; the computation is the imaginary part of Theorem 4.1. ∎

Theorem 1.3 (Gauss's law propagates). Let $G \equiv \nabla\cdot\mathbf{E} - \rho_b + e\rho$. For every solution of (M) and (A), $\partial_0 G = 0$. Hence (G) holds for all times if it holds at one time.

Proof. $\partial_0\nabla\cdot\mathbf{E} = \nabla\cdot\big(\nabla^2\mathbf{A} - \nabla\nabla\cdot\mathbf{A}\big) + e\nabla\cdot(\rho\mathcal{V}) = 0 - e\,\partial_0\rho$ by (C). ∎

This is the content of (A5)–(A7). The source then sets the "time-independent term" to zero, which is the choice of imposing (G) at $x^0 = 0$.

Theorem 1.4 (The pair $A^0 = 0$, $\nabla\cdot\mathbf{A} = 0$ freezes the density). Suppose $A^0 \equiv 0$ and $\nabla\cdot\mathbf{A} \equiv 0$. Then $\nabla\cdot\mathbf{E} \equiv 0$, hence

$$\partial_0\rho \equiv 0,$$

and if (G) holds, $e\rho \equiv \rho_b$. In particular, with $\rho_b = 0$ and (G), $\varphi \equiv 0$.

Proof. $\nabla\cdot\mathbf{E} = \partial_0\nabla\cdot\mathbf{A} - \nabla^2 A^0 = 0$. Theorem 1.3 gives $\partial_0(e\rho) = -\partial_0 G = 0$. ∎

The source calls this pair "the Coulomb gauge" (before (A2)). It is not a gauge: a gauge choice removes one function, and this pair removes two. It is a restriction on solutions, and Theorem 1.4 says which: those with time-independent density.

Theorem 1.5 (Coulomb gauge, consistent form). On $\mathbb{R}^3$ with decaying fields impose $\nabla\cdot\mathbf{A} = 0$ and define A^0 by the Poisson equation

$$-\nabla^2 A^0 = \rho_b - e\rho, \qquad A^0 = \frac{1}{4\pi}\int \frac{\rho_b(\mathbf{y}) - e\rho(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}\,d^3y. \tag{1.2}$$

Then (G) holds identically, the longitudinal part of (A) is equivalent to (C), and the transverse part of (A) is

$$\partial_0^2\mathbf{A} = \nabla^2\mathbf{A} + (e\rho\,\mathcal{V})_T\,, \tag{1.3}$$

where $(\cdot)_T$ is the divergence-free projection. The system (M), (1.2), (1.3) is the model with no restriction on solutions.

Proof. With $\nabla\cdot\mathbf{A} = 0$, $\nabla\cdot\mathbf{E} = -\nabla^2 A^0$, so (1.2) is (G). Take the divergence of (A): $-\partial_0\nabla^2 A^0 = e\nabla\cdot(\rho\mathcal{V})$, which by (1.2) is $\partial_0(\rho_b - e\rho) = e\nabla\cdot(\rho\mathcal{V})$, i.e. (C). The transverse part of (A) with $\mathbf{E} = \partial_0\mathbf{A} - \nabla A^0$ is (1.3). ∎

On a periodic box, (1.2) is solvable only if $\int(\rho_b - e\rho) = 0$: the box must be neutral. With $\rho_b = 0$ this forces $N = 0$. A neutralizing background is therefore not optional on a torus.

Proposition 1.6 (The price of the repair). Under Theorem 1.5, equation (M) carries the term eA^0 with A^0 given by (1.2). Let $\delta \equiv \sup|\rho_b - e\rho|$ on the support.

a. Slab: if $\rho_b - e\rho$ depends on x^3 only and vanishes outside $|x^3| \le L_3/2$, then A^0, normalized to vanish on the two faces, satisfies $\sup|A^0| \le \delta L_3^2/8$.

b. Ball: if it vanishes outside a ball of radius a, then $\sup|A^0| \le \delta a^2/2$.

An additive constant in A^0 shifts every level equally; the oscillation of A^0 over the support is what shifts them differently, and it is bounded by the same numbers. Consequently the single-particle levels used in Section 2 are shifted relative to one another by at most $e\sup|A^0|$, and the two-level reduction there requires

$$e \sup |A^0| \ll \Omega. \qquad (1.4)$$

With $\rho_b = 0$ (the source) and $\rho \approx \bar{n}$, (a) reads $e^2 \bar{n} L_3^2/8 = m\,\omega_p^2 L_3^2/8 \ll \Omega$.

Proof. (a) The one-dimensional problem $-f'' = q$ on $[-L/2, L/2]$ with $|q| \le \delta$ and $f(\pm L/2) = 0$ satisfies $|f| \le \delta(L^2/4 - x^2)/2 \le \delta L^2/8$ by comparison with $q \equiv \pm\delta$. (b) $|A^0(\mathbf{x})| \le \frac{\delta}{4\pi}\int_{|\mathbf{y}|\le a} \frac{d^3y}{|\mathbf{x}-\mathbf{y}|} \le \frac{\delta}{4\pi}\int_{|\mathbf{y}|\le a} \frac{d^3y}{|\mathbf{y}|} = \delta a^2/2$ (the integral is maximal at the centre). ■

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2. Two-level emitter and the pulse

Hypotheses.

- **H2.1 (two levels).** $-\nabla^2/2m + U$ has real normalized eigenfunctions ϕ_0, ϕ_1 with eigenvalues $E_0 < E_1$; $\Omega \equiv E_1 - E_0$; and $\varphi(x) = \sum_{n=0,1} e^{-iE_n x^0}\chi_n(x^0)\phi_n(\mathbf{x})$ with $|\chi_0|^2 + |\chi_1|^2 = N$. On a set where $\phi_0 \neq 0$, the ratio ϕ_1/ϕ_0 takes at least three values.
- **H2.2 (emission geometry).** $\mathbf{A} = A_3(x^0, x^1, x^2)\,\hat{\mathbf{e}}_3$ and $A^0 = 0$. (This is the pair of Theorem 1.4; see Theorem 2.9.)
- **H2.3 (resonant shell).** $A_3 = \int \frac{d^2k}{(2\pi)^2}\,\frac{1}{2}\left[\alpha_{\mathbf{k}} e^{-i(\omega_k x^0 - \mathbf{k}\cdot\mathbf{x})} + \text{c.c.}\right]$ with $\mathbf{k} = (k^1, k^2, 0)$ and $\omega_k^2 = |\mathbf{k}|^2 + \omega_p^2$; only the modes with $\omega_k = \Omega$ are kept. This requires

  $$\Omega^2 \ge \omega_p^2. \qquad (2.0)$$

  The shell measure is $\Omega^2/4\pi$ (this is (A29)); $\mathcal{A} \equiv \int_{\omega_k=\Omega} \alpha_{\mathbf{k}}$, made real by a constant phase.
- **H2.4 (slow envelopes).** $|\partial_0 \chi_n| \le \varepsilon\,\Omega\,|\chi_n|$ with $\varepsilon < \frac{1}{2}$, and $|\partial_0 \alpha_{\mathbf{k}}| \ll \Omega |\alpha_{\mathbf{k}}|$.
- **H2.5 (dipole).** $e^{i\mathbf{k}\cdot\mathbf{x}} \simeq 1$ on the support of ϕ_n: $L_1, L_2 \ll 2\pi/\sqrt{\Omega^2 - \omega_p^2}$.

The source states (2.0) as $\Omega \ge e^2|\bar{\varphi}|^2/m$ (after (A29) and (A47)); the two sides have different dimensions, and (2.0) is the condition under which the shell in (A29) exists.

Definition 2.2 (Reduced model). Let $d \equiv \frac{1}{m}\int \phi_0\,\partial_3\phi_1 \in \mathbb{R}$, $g \equiv ed/2$, $\kappa \equiv \dfrac{g\Omega}{4\pi L_3}$. The reduced model is

$$\dot{\chi}_0 = g\mathcal{A}\chi_1, \qquad \dot{\chi}_1 = -g\mathcal{A}\chi_0, \qquad \dot{\mathcal{A}} = \kappa R, \qquad R \equiv 2\,\mathrm{Re}(\chi_0^*\chi_1). \qquad (2.1)$$

Under H2.1–H2.5, equations (M) and (A) reduce to (2.1); this is (A14)–(A35) with the misprint in (A31) (χ_1 for χ_0) corrected and the phase of $\mathcal{A}$ fixed. We take (2.1) as the definition and prove everything below from it.

Proposition 2.3 (Invariants). Along (2.1), with $Z \equiv |\chi_1|^2 - |\chi_0|^2$,

$$N = |\chi_0|^2 + |\chi_1|^2, \qquad I \equiv \mathrm{Im}(\chi_0^*\chi_1), \qquad \mathcal{E} \equiv \tfrac{1}{2}\mathcal{A}^2 + \frac{\Omega}{8\pi L_3} Z$$

are constant, and $Z^2 + R^2 = N^2 - 4I^2$.

Proof. $\dot{Z} = -2g\mathcal{A}R$, $\dot{R} = 2g\mathcal{A}Z$, $\frac{d}{dt}(\chi_0^*\chi_1) = g\mathcal{A}(|\chi_1|^2 - |\chi_0|^2) = g\mathcal{A}Z$ is real, so $\dot{I} = 0$; $\dot{N} = 2\,\mathrm{Re}(\chi_0^*\dot{\chi}_0 + \chi_1^*\dot{\chi}_1) = 2g\mathcal{A}\,\mathrm{Re}(\chi_0^*\chi_1 - \chi_1^*\chi_0) = 0$; $\dot{\mathcal{E}} = \kappa\mathcal{A}R - \frac{\Omega}{8\pi L_3}2g\mathcal{A}R = 0$. Finally $R^2 + 4I^2 = 4|\chi_0|^2|\chi_1|^2$ and $Z^2 + 4|\chi_0|^2|\chi_1|^2 = N^2$. ■

Put $N' \equiv \sqrt{N^2 - 4I^2}$ and, for $N' > 0$, define θ by $Z = N'\cos\theta$, $R = N'\sin\theta$. Then $\dot{\theta} = 2g\mathcal{A}$, which is (A40). For real initial data $I = 0$ and $N' = N$.

Theorem 2.4 (Pendulum). θ obeys

$$\ddot{\theta} = \omega_c^2 \sin\theta, \qquad \omega_c^2 \equiv 2g\kappa N' = \frac{g^2\Omega N'}{2\pi L_3}. \qquad (2.2)$$

Equivalently $\psi \equiv \theta - \pi$ is a pendulum, $\ddot{\psi} = -\omega_c^2\sin\psi$. The state $\theta = 0$ (all quanta in the upper level, $Z = N'$) is the unstable equilibrium; $\theta = \pi$ is the stable one.

Proof. $\ddot\theta = 2g\dot{\mathcal{A}} = 2g\kappa R = 2g\kappa N' \sin\theta$. ∎

Theorem 2.5 (Lossless dynamics: no emission, a periodic train). Let $\mathcal{A}(0) = 0$ and $\theta(0) = \theta_0 \in (0, \pi)$. Then

i. θ oscillates between θ_0 and $2\pi - \theta_0$ with period

$$T(\theta_0) = \frac{4}{\omega_c} K\!\left(\cos\tfrac{\theta_0}{2}\right), \qquad K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2\phi}}; \tag{2.3}$$

ii. $\mathcal{A}(t) = \dfrac{\dot\theta}{2g}$ satisfies $|\mathcal{A}| \le \dfrac{\omega_c}{g}\cos\dfrac{\theta_0}{2}$, with equality at $\theta = \pi$, and $\mathcal{A}$ returns to 0 every half period;

iii. as $\theta_0 \downarrow 0$, $T(\theta_0) = \dfrac{4}{\omega_c}\ln\dfrac{8}{\theta_0} + o(1) \to \infty$, and each pulse of the train converges to

$$\mathcal{A}(t) = \frac{\omega_c}{g}\operatorname{sech}\big(\omega_c(t - t_1)\big). \tag{2.4}$$

Proof. Multiply (2.2) by $\dot\theta$: $\tfrac12\dot\theta^2 + \omega_c^2\cos\theta = \omega_c^2\cos\theta_0$. The potential $\omega_c^2\cos\theta$ has its maximum ω_c^2 at $\theta = 0$, so the orbit is a libration about $\theta = \pi$ with turning points where $\cos\theta = \cos\theta_0$, i.e. θ_0 and $2\pi - \theta_0$. In the variable $\psi = \theta - \pi$ this is the pendulum of amplitude $\pi - \theta_0$, whose period is $\frac{4}{\omega_c}K(\sin\frac{\pi-\theta_0}{2}) = \frac{4}{\omega_c}K(\cos\frac{\theta_0}{2})$. At $\theta = \pi$, $\dot\theta^2 = 2\omega_c^2(1 + \cos\theta_0) = 4\omega_c^2\cos^2\frac{\theta_0}{2}$, giving (ii). For (iii), $K(k) = \ln\frac{4}{\sqrt{1-k^2}} + o(1)$ as $k \uparrow 1$ and $\sqrt{1 - \cos^2\frac{\theta_0}{2}} = \sin\frac{\theta_0}{2}$; on the separatrix ($\theta_0 = 0$) the energy integral reads $\dot\theta = 2\omega_c\sin\frac{\theta}{2}$, whose solution is $\tan\frac{\theta}{4} = e^{\omega_c(t-t_1)}$, and $\sin\frac{\theta}{2} = \operatorname{sech}\,\omega_c(t - t_1)$. ∎

So the lossless reduced model has a hyperbolic-secant pulse, but as the limit of a periodic train whose period diverges logarithmically in the seed; and $\mathcal{E}$ is conserved, so nothing is emitted: the field returns its energy to the matter every half period. A single emitted pulse requires an energy sink.

H2.6 (loss). Radiation leaves the slab at rate $1/L_3$:

$$\dot{\mathcal{A}} = \kappa R - \frac{\mathcal{A}}{L_3}. \tag{2.5}$$

This is (A42). It is a model of escape, not a consequence of (1.1).

Theorem 2.6 (Overdamped limit: the single pulse). Put $\tau_R \equiv \dfrac{2\pi}{g^2\Omega N'}$, so that $\omega_c^2 = \dfrac{1}{\tau_R L_3}$. If

$$L_3 \ll \tau_R, \tag{2.6}$$

then on times $x^0 \gg L_3$ the solution of (2.1) with (2.5) is, to leading order in L_3/τ_R,

$$\mathcal{A} = \kappa L_3 N' \sin\theta, \qquad \dot\theta = \frac{\sin\theta}{\tau_R}, \qquad \theta(x^0) = 2\arctan\!\left(e^{x^0/\tau_R}\tan\frac{\theta_0}{2}\right), \tag{2.7}$$

$$\mathcal{A}(x^0) = \frac{g\Omega N'}{4\pi}\operatorname{sech}\frac{x^0 - \tau_0}{\tau_R}, \qquad \tau_0 = -\tau_R\ln\tan\frac{\theta_0}{2}. \tag{2.8}$$

Here θ increases monotonically from θ_0 to π: the pulse is emitted once and the energy $\mathcal{E}$ decreases by the amount that leaves through the loss term.

Proof. With $\dot{\mathcal{A}} = O(\mathcal{A}/\tau_R)$ and $1/L_3 \gg 1/\tau_R$, (2.5) gives $\mathcal{A} = \kappa L_3 N' \sin\theta\,(1 + O(L_3/\tau_R))$. Then $\dot\theta = 2g\mathcal{A} = 2g\kappa L_3 N'\sin\theta = \sin\theta/\tau_R$, which integrates to (2.7); differentiating, $\mathcal{A} = \dot\theta/2g = \frac{1}{2g\tau_R}\sin\theta$ and $\sin\theta = \operatorname{sech}((x^0 - \tau_0)/\tau_R)$ with τ_0 as stated. Monotonicity: $\sin\theta > 0$ on $(0,\pi)$. ∎

(2.8) is (A45)–(A47). The width τ_R is independent of L_3 only inside the regime (2.6); outside it the width is $1/\omega_c = \sqrt{\tau_R L_3}$ of Theorem 2.5. Both regimes have the shape sech; the shape does not identify the regime, the width does.

Corollary 2.7 (Every pulse needs a seed). $\theta_0 = 0$, $\mathcal{A} = 0$ is a stationary solution of (2.1), of (2.1)+(2.5), and of (2.7). The delay $\tau_0 \to +\infty$ and the period $T \to \infty$ as $\theta_0 \downarrow 0$. No equation of Sections 1–2 produces $\theta_0 \neq 0$; it is a hypothesis.

H2.7 (seed). $\theta_0 \in (0, \pi)$ is given.

The source uses θ_0 in (A41), (A44), (A47) and writes "definition of θ_0 is in Appendix A"; no definition is there.

Proposition 2.8 (A frozen density has no coherence). Assume H2.1 and H2.4. If $\partial_0 \rho \equiv 0$, then $\chi_0^* \chi_1 \equiv 0$; hence $R \equiv 0$, $\mathcal{A}$ is constant, and no pulse exists.

Proof. With real ϕ_n,

$$\rho = |\chi_0|^2 \phi_0^2 + |\chi_1|^2 \phi_1^2 + 2\phi_0 \phi_1 \,\mathrm{Re}\!\left(\chi_0^* \chi_1 e^{-i\Omega x^0}\right).$$

Dividing by ϕ_0^2 on the set of H2.1, ρ/ϕ_0^2 is a quadratic polynomial in $s = \phi_1/\phi_0$ with coefficients $|\chi_0|^2$, $2\,\mathrm{Re}(\chi_0^* \chi_1 e^{-i\Omega x^0})$, $|\chi_1|^2$; a quadratic constant in time at three values of s has time-independent coefficients. So $|\chi_0|$, $|\chi_1|$ are constant and $\mathrm{Re}(c\, e^{-i\Omega x^0})$ is constant, where $c \equiv \chi_0^* \chi_1$ has constant modulus. If $c \neq 0$ write $c = |c| e^{i\eta}$: $\cos(\eta - \Omega x^0)$ constant forces $\dot\eta = \Omega$, so $|\dot c| = \Omega |c|$. But $|\dot c| \le |\dot\chi_0||\chi_1| + |\chi_0||\dot\chi_1| \le 2\varepsilon\Omega |c| < \Omega |c|$ by H2.4. Hence $c = 0$. ∎

Theorem 2.9 (Appendix A, as written, has no pulse; the repair and its price). H2.2 places the system under Theorem 1.4 with $\rho_b = 0$: the density is frozen. By Proposition 2.8 the coherence vanishes and (2.8) cannot occur. The pulse (2.8) is a theorem of the reduced model (2.1)+(2.5)+H2.7. The reduced model follows from (1.1) only after replacing H2.2 by Theorem 1.5 (Coulomb gauge with A^0 from (1.2)), and then only under the smallness condition (1.4), which with $\rho_b = 0$ is $e^2 \bar n L_3^2/8 \ll \Omega$.

Proof. Theorems 1.4, 1.5, Propositions 1.6, 2.8. ∎

Proposition 2.10 (Transverse current). H2.2 also sets the x^1, x^2 components of (A) to zero, which by (A) requires $\rho\, \partial_{1,2}\beta = 0$. Under H2.1 and H2.4, if ϕ_1/ϕ_0 depends on x^1 or x^2, this forces $\chi_0^* \chi_1 = 0$.

Proof. With $s = \phi_1/\phi_0$ and $w = e^{-i\Omega x^0}$, $\beta = -E_0 x^0 + \arg(\chi_0 + w\chi_1 s)$ up to a constant, so $\partial_{1,2}\beta = \mathrm{Im}(w\chi_0^*\chi_1)\,\partial_{1,2} s/|\chi_0 + w\chi_1 s|^2$. If $\partial_{1,2} s \neq 0$ somewhere, $\mathrm{Im}(e^{-i\Omega x^0}\chi_0^*\chi_1) \equiv 0$ there, and the argument of Proposition 2.8 gives $\chi_0^*\chi_1 = 0$. ∎

The source's hypothesis that ∇_3 dominates ((A10)) is the case $\phi_1/\phi_0 = f(x^3)$, in which the transverse current vanishes identically and only Theorem 2.9 remains.

Proposition 2.11 (What decides whether a pulse occurs). The reduced model has the parameters $g, \Omega, N', L_3, \theta_0$ and, for the pulse to be observed before the phases of the N' emitters randomize, a dephasing time T_2 with

$$\tau_R = \frac{2\pi}{g^2 \Omega N'} < T_2, \qquad L_3 \ll \tau_R, \qquad \Omega^2 \geq \omega_p^2, \qquad e^2 \bar n L_3^2 \ll 8\Omega. \tag{2.9}$$

The source gives no values for g, N', L_3, θ_0 or T_2, and for Ω only the order of magnitude of a water rotational spacing (4 meV, its Section 1). Whether (2.9) holds in any medium is therefore open, and the source's own discussion ("the balance of decoherence and error corrections … in the open system") names the computation it does not perform.

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3. Screened waves, holographic readout, capacity

Hypotheses.

- **H3.1 (static neutral medium).** $\rho = \rho(\mathbf{x})$, $\mathcal{V} \equiv 0$, $\rho_b = e\rho$. Then Theorem 1.4 permits $A^0 = 0$, $\nabla \cdot \mathbf{A} = 0$ consistently: a static medium is the one case in which the pair of H2.2 is harmless.
- **H3.2 (screening and source).** Equation (A) acquires a term $-M^2 \mathbf{A}$ with a supplied constant $M^2 \geq 0$, and a supplied external field enters as $M^2 \tilde{\mathbf{A}}$.
- **H3.3 (polarization).** $\mathbf{A} = A_3 \hat{\mathbf{e}}_3$, $\tilde{\mathbf{A}} = \tilde A_3 \hat{\mathbf{e}}_3$, both independent of x^3, so $\nabla \cdot \mathbf{A} = 0$.

Definition 3.1 (Screened wave equation).

$$\left[\partial_0^2 - \nabla^2 + \omega_p^2(\mathbf{x}) + M^2\right] A_3 = M^2 \tilde A_3, \qquad \omega_p^2(\mathbf{x}) \equiv \frac{e^2 \rho(\mathbf{x})}{m}. \tag{3.1}$$

This is (A48) with its remainder $O(\partial\varphi)$ equal to zero under H3.1.

Theorem 3.2 (Transmittance). Let $\tilde{A}_3 = b\, e^{i(k^0 x^0 - \mathbf{k}\cdot\mathbf{x})}$ with $k^0 = |\mathbf{k}|$ and $\mathbf{k} \perp \hat{\mathbf{e}}_3$. Put

$$T(\mathbf{x}) \equiv \frac{M^2}{M^2 + \omega_p^2(\mathbf{x})}. \qquad (3.2)$$

Then $A_3 = T\tilde{A}_3$ satisfies (3.1) with the source replaced by $M^2\tilde{A}_3 + \mathcal{R}$, where

$$\mathcal{R} = \left(2i\,\mathbf{k}\cdot\nabla T - \nabla^2 T\right)\tilde{A}_3. \qquad (3.3)$$

Proof. $(\partial_0^2 - \nabla^2)(T\tilde{A}_3) = T(-k_0^2 + |\mathbf{k}|^2)\tilde{A}_3 - 2\nabla T\cdot\nabla\tilde{A}_3 - (\nabla^2 T)\tilde{A}_3 = 2i(\mathbf{k}\cdot\nabla T)\tilde{A}_3 - (\nabla^2 T)\tilde{A}_3$, and $(\omega_p^2 + M^2)T = M^2$. ∎

$\mathcal{R}$ is small when T varies slowly on the scale $1/|\mathbf{k}|$; (A49) is Theorem 3.2 with $\mathcal{R}$ dropped.

Corollary 3.3 (The medium is seen only through s). With $s(\mathbf{x}) \equiv \omega_p^2(\mathbf{x})/M^2$: if $M^2 = 0$ then $T \equiv 0$ and $\mathcal{R} \equiv 0$, so the external wave does not enter the medium at all; as $M^2 \to \infty$, $T \to 1$; and for $s < 1$,

$$T = 1 - s + s^2 - \cdots. \qquad (3.4)$$

H3.4 (recorded pattern). On the recording plane, $\rho(x^1, x^2) = c\,|r(x^1, x^2) + o(x^1, x^2)|^2$ with $c > 0$, r a reference wave and o an object wave, and $\tilde{A}_3 = r$.

Theorem 3.4 (Reconstruction identity). Under H3.1–H3.4, to first order in s,

$$A_3 - \tilde{A}_3 = -\sigma\Big[\left(|r|^2 + |o|^2\right) r \;+\; |r|^2\, o \;+\; r^2\, o^*\Big] + O(s^2), \qquad \sigma \equiv \frac{c\,e^2}{mM^2}. \qquad (3.5)$$

The third term is the object wave multiplied by $|r|^2$; the fourth is its conjugate (the twin image); and the third has amplitude at most $s_r|o|$ with $s_r \equiv \sigma \sup|r|^2$.

Proof. (3.4) gives $A_3 - \tilde{A}_3 = -s\,r + O(s^2)$ with $s = \sigma|r+o|^2 = \sigma(|r|^2 + |o|^2 + r^* o + r o^*)$; multiply by r. ∎

This is (A52), which prints $\tilde{A}_i$ on the left for A_i. Nothing selects the third term over the fourth; that is the ordinary twin-image problem of in-line holography.

Corollary 3.5 (Numbers). At the source's Section 4 values, $e = 0.3$, $\rho = m^3$ (its $|\bar{\varphi}|^2 = 1$ in units of m) and $M^2 = 2m^2$: $s = e^2\rho/(mM^2) = 0.045$. The reconstructed object wave carries at most $s_r \approx 0.045$ of the object's amplitude: 4.5%, at the only parameter values the source gives.

Remark 3.6 (No write). In Definition 3.1 the density ρ is data. No equation of Sections 1–3 changes ρ (H3.1 sets $\mathcal{V} = 0$), and no equation contains o except through H3.4. The pattern $|r + o|^2$ is assumed, not produced, and o has no source.

Lemma 3.7 (Cell count). Partition a region of area S and thickness d into cells of lateral side $\ell_\parallel$ and depth $\ell_\perp$. The number of cells is

$$C = \frac{S\,d}{\ell_\parallel^2\,\ell_\perp}. \qquad (3.6)$$

With the convention $\ell_\parallel = \lambda/2$, $\ell_\perp = \lambda/n_r$ (the source's, after its reference for the 2-dimensional limit $4/\lambda^2$), $C_3 = 4n_r S d/\lambda^3$, and the surface alone holds $C_2 = 4S/\lambda^2$; their ratio is

$$\frac{C_3}{C_2} = \frac{n_r d}{\lambda}. \qquad (3.7)$$

Proposition 3.8 (The petabyte is the thickness factor). With $n_r = 1.3$, $d = 2\,\mathrm{mm}$, $\lambda = 500\,\mathrm{nm}$, $S = 2500\,\mathrm{cm}^2$:

$$C_2 = 4.0 \times 10^{12}\ (0.5\ \mathrm{TB}), \qquad \frac{n_r d}{\lambda} = 5200, \qquad C_3 = 2.08 \times 10^{16}\ (2.6\ \mathrm{PB}).$$

At $\lambda = 1500\,\mathrm{nm}$, $C_3 = 0.096$ PB; at $\lambda = 310\,\mu\mathrm{m}$ (the wavelength of the $4\ \mathrm{meV}$ scale the source quotes for water), $C_3 = 8.7 \times 10^7$ bits (11 MB). The comparison figure 250×10^{12} synapses $\times\ 1$ bit is 31 TB; at 4.7 bits per synapse it is 147 TB.

One cell is one bit only under the hypothesis **H3.5**: a write and a read exist that set and recover one bit per cell. Neither is given (Remark 3.6).

Lemma 3.9 (Erasure bound versus retention). The Landauer bound $k_B T \ln 2$ is a lower bound on the heat released by one logically irreversible erasure. It is not a bound on how long a stored value persists. For a two-state cell separated by a barrier ΔE with attempt time τ_a, the Arrhenius lifetime is $\tau = \tau_a e^{\Delta E / k_B T}$, so retention for a time τ_{ret} needs

$$\Delta E \geq k_B T \ln \frac{\tau_{\mathrm{ret}}}{\tau_a}. \qquad (3.8)$$

Proposition 3.10 (Numbers). At $T = 310\,\mathrm{K}$, $k_B T = 26.7\ \mathrm{meV}$ and $k_B T \ln 2 = 18.5\ \mathrm{meV}$ (the source rounds to 20). With $\tau_a = 1\ \mathrm{ps}$: $\tau_{\mathrm{ret}} = 1\ \mathrm{s}$ needs $\Delta E \geq 0.74\ \mathrm{eV}$; 1 hour, $0.96\ \mathrm{eV}$; 1 year, $1.20\ \mathrm{eV}$. The photon energy $2.4\ \mathrm{eV}$ at $500\ \mathrm{nm}$ is not a barrier of the medium; no barrier is given. The "maximum wavelength $60\ \mu\mathrm{m}$" is $hc/(k_B T \ln 2)$: $67\ \mu\mathrm{m}$, or $62\ \mu\mathrm{m}$ with the rounded 20 meV. It is a conversion of units, not a property of the medium.

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4. Driven density patterns: the classical sector

Theorem 4.1 (Polar form). On any region where $\rho > 0$, equation (M) is equivalent to the pair

$$\text{(C)} \quad \partial_0 \rho = -\nabla \cdot (\rho \mathcal{V}), \qquad \text{(B)} \quad eA^0 - \partial_0 \beta = -\frac{\nabla^2 \sqrt{\rho}}{2m\sqrt{\rho}} + \frac{m}{2}\mathcal{V}^2 + U.$$

Proof. With $D \equiv \nabla - ie\mathbf{A}$ and $\varphi = \sqrt{\rho} e^{i\beta}$: $D\varphi = e^{i\beta}(\nabla\sqrt{\rho} + im\sqrt{\rho}\,\mathcal{V})$ and

$$D^2\varphi = e^{i\beta}\left[\nabla^2\sqrt{\rho} - m^2\mathcal{V}^2\sqrt{\rho} + im\left(\nabla\cdot(\sqrt{\rho}\,\mathcal{V}) + \mathcal{V}\cdot\nabla\sqrt{\rho}\right)\right].$$

(M) reads $e^{i\beta}[i\partial_0\sqrt{\rho} - \sqrt{\rho}\,\partial_0\beta + eA^0\sqrt{\rho}] = -\frac{1}{2m}D^2\varphi + U\sqrt{\rho}e^{i\beta}$. The imaginary part is $\partial_0\sqrt{\rho} = -\frac{1}{2}(\nabla\cdot(\sqrt{\rho}\mathcal{V}) + \mathcal{V}\cdot\nabla\sqrt{\rho})$, which times $2\sqrt{\rho}$ is (C). The real part divided by $\sqrt{\rho}$ is (B). ■

B. is (A70) with the sign of the $\mathcal{V}^2$ term corrected: the source prints $-\frac{e}{2m}(A_i - \partial_i\beta/e)^2$, while (A81) and (A87), and the derivation, carry $+$. A free plane wave satisfies (B) and not the printed form.

Lemma 4.2 (Unitary gauge). If $\rho > 0$ everywhere, the gauge transformation $\varphi \to e^{-i\beta}\varphi$, $A^0 \to A^0 - \partial_0\beta/e$, $\mathbf{A} \to \mathbf{A} - \nabla\beta/e$ sets $\beta \equiv 0$. In this gauge $\mathcal{V} = -e\mathbf{A}/m$ and (B) determines A^0:

$$eA^0 = -\frac{\nabla^2\sqrt{\rho}}{2m\sqrt{\rho}} + \frac{e^2}{2m}\mathbf{A}^2 + U. \qquad (4.1)$$

This is the source's "we omit terms $-\partial_\mu\beta/e$" (after (A81)); it is not an approximation but a gauge choice, and it is incompatible with any further condition on A^0 or $\nabla\cdot\mathbf{A}$. (4.1) with $U = 0$ is (A87).

Hypotheses.

- **H4.1 (planar).** $A_3 = 0$, nothing depends on x^3, $U = 0$.
- **H4.2 (supplied response).** The matter's back-reaction beyond mean field is replaced by $-M^2\mathbf{A} - \gamma\,\partial_0\mathbf{A}$ with supplied constants $M^2 \geq 0, \gamma \geq 0$. (This is (A75); the derivation from the self-energy is not reproduced here, and the constants are taken as data.)
- **H4.3 (layers).** Copies $n = 1, \dots, N$ of the fields, coupled by $v(\mathbf{A}^{(n-1)} + \mathbf{A}^{(n+1)})$ with a supplied constant v, a supplied drive $\mathbf{A}^{(0)}$, and $\mathbf{A}^{(N+1)} \equiv 0$.
- **H4.4 (constraint).** $A^{0(n)}$ is given by (4.1); equation (G) is not imposed.

Definition 4.3 (Layer system). For $n = 1, \dots, N$, with $E_i^{(n)} \equiv \partial_0 A_i^{(n)} - \partial_i A^{0(n)}$:

$$\partial_0 E_1^{(n)} = \partial_2^2 A_1^{(n)} - \partial_1\partial_2 A_2^{(n)} - \frac{e^2\rho^{(n)}}{m}A_1^{(n)} - M^2 A_1^{(n)} - \gamma\,\partial_0 A_1^{(n)} + v\big(A_1^{(n-1)} + A_1^{(n+1)}\big),$$

$$\partial_0 E_2^{(n)} = \partial_1^2 A_2^{(n)} - \partial_1\partial_2 A_1^{(n)} - \frac{e^2\rho^{(n)}}{m}A_2^{(n)} - M^2 A_2^{(n)} - \gamma\,\partial_0 A_2^{(n)} + v\big(A_2^{(n-1)} + A_2^{(n+1)}\big),$$

$$\partial_0 \rho^{(n)} = \frac{e}{m}\nabla\cdot\big(\rho^{(n)}\mathbf{A}^{(n)}\big), \qquad eA^{0(n)} = -\frac{\nabla^2\sqrt{\rho^{(n)}}}{2m\sqrt{\rho^{(n)}}} + \frac{e^2}{2m}\big(\mathbf{A}^{(n)}\big)^2. \tag{4.2}$$

This is (A82)–(A87). The source prints $e\rho/m$ for $e^2\rho/m$ in (A76) and (A78).

Proposition 4.4 (The one quantum term). Restoring $\hbar$, (4.1) reads $eA^0 = -\dfrac{\hbar^2\nabla^2\sqrt{\rho}}{2m\sqrt{\rho}} + \dfrac{e^2}{2m}\mathbf{A}^2$. Every other term of (4.2) is independent of $\hbar$. Setting that term to zero leaves a classical charged fluid: continuity, Ampère's law with the supplied M^2, γ, v, and the constraint $eA^0 = e^2\mathbf{A}^2/2m$.

Proof. Dimensional restoration of (B): $\mathcal{V} = (\hbar\nabla\beta - e\mathbf{A})/m$ and the quantum pressure carries $\hbar^2$; in the unitary gauge no other $\hbar$ remains. ∎

The source reports that this term produces "slight distortions" of Figure 6f and "does not seriously affect" the result. That is a statement that the simulation is insensitive to its only quantum term.

Theorem 4.5 (Gauss's law drifts). For a solution of (4.2) put $G^{(n)} \equiv \nabla\cdot\mathbf{E}^{(n)} - \rho_b + e\rho^{(n)}$. Then

$$\partial_0 G^{(n)} = -\left(M^2 + \gamma\,\partial_0\right)\nabla\cdot\mathbf{A}^{(n)} + v\,\nabla\cdot\big(\mathbf{A}^{(n-1)} + \mathbf{A}^{(n+1)}\big). \tag{4.3}$$

In particular $G^{(n)}$ is conserved if and only if the right side vanishes; for $M^2 = \gamma = v = 0$ it is conserved (Theorem 1.3).

Proof. Take the divergence of the first two equations of (4.2): $\nabla\cdot(\nabla^2\mathbf{A} - \nabla\nabla\cdot\mathbf{A}) = 0$; $-\frac{e^2}{m}\nabla\cdot(\rho\mathbf{A}) = -e\,\partial_0\rho$ by the third; $\nabla\cdot(\partial_0\mathbf{A}) = \partial_0\nabla\cdot\mathbf{A}$; the rest is (4.3). ∎

For the targets (4.5) below, $\nabla\cdot\mathbf{A}_{\mathrm{tgt}} \neq 0$ on the interior, so $G^{(N)}$ drifts. A rerun of the simulation at the source's parameters (reported in the companion checks of this folder, not replayed here) finds $|\partial_0 G^{(N)}|$ between 1.1 and 2.3 times $|\partial_0(e\rho^{(N)})|$ over the run: the quantity the simulation reports as the hologram and the violation of Gauss's law are of the same size. The source reports conservation of $\int\rho^{(N)}$ to $10^{-6}\,\%$; that is the third equation of (4.2) on a periodic box and is unrelated to (4.3).

Theorem 4.6 (Inverse design is a Chebyshev recursion). Drop time derivatives and the $e^2\rho/m$ term in (4.2) (as the source does), write $u^{(n)}$ for the static fields, and impose $u^{(N)} = \mathbf{A}_{\mathrm{tgt}}$, $u^{(N+1)} = 0$. Then

$$u_1^{(n-1)} = \frac{M^2u_1^{(n)} - \partial_2^2u_1^{(n)} + \partial_1\partial_2u_2^{(n)}}{v} - u_1^{(n+1)}, \qquad u_2^{(n-1)} = \frac{M^2u_2^{(n)} - \partial_1^2u_2^{(n)} + \partial_1\partial_2u_1^{(n)}}{v} - u_2^{(n+1)}. \tag{4.4}$$

If $u_1^{(N)} = u_1(x^1)$ and $u_2^{(N)} = u_2(x^2)$, all derivative terms vanish and

$$u^{(N-J)} = U_J(c)\,u^{(N)}, \qquad c \equiv \frac{M^2}{2v}, \qquad U_0 = 1,\ U_1 = 2c,\ U_{J+1} = 2cU_J - U_{J-1}, \tag{4.5}$$

with U_J the Chebyshev polynomial of the second kind; for $c > 1$, $U_J(c) = \dfrac{\sinh((J+1)\eta)}{\sinh\eta}$, $c = \cosh\eta$.

Proof. The static, drive-free first equation of (4.2) at layer n is $0 = \partial_2^2u_1^{(n)} - \partial_1\partial_2u_2^{(n)} - M^2u_1^{(n)} + v(u_1^{(n-1)} + u_1^{(n+1)})$; solve for $u_1^{(n-1)}$. For separable targets the recursion is $u^{(n-1)} + u^{(n+1)} = 2c\,u^{(n)}$ with $u^{(N+1)} = 0$, which is the defining recursion of U_J in $J = N - n$. The closed form is the standard one. ∎

The source's (A88) prints $-\partial_1^2u_1$ where (4.4) has $-\partial_2^2u_1$ (and symmetrically in (A89)); for its targets the difference is invisible because both vanish.

Corollary 4.7 (The input amplitude). With $N = 4$, $M^2/v = 2.5$ ($c = 1.25$) and target amplitude 0.1: $U_4(1.25) = 21.3125$, so $\max|u^{(0)}| = 2.13$. The source's "$0.1 \times (M^2/m^2)^4/v^N = 3.9$" is the recursion without the term $-u^{(n+1)}$; its Figure 4 shows "approximately 3". Keeping the $e^2\rho/m$ term, $c = (M^2 + \omega_p^2)/2v = 1.30625$, $U_4 = 27.11$ and $\max|u^{(0)}| = 2.71$.

Targets (the source's (10)–(11), in lattice units $x^i = n^i a_s$, $m a_s = 1$, $N_s = 64$, $N_1 = 30$, $N_2 = 20$):

$$A_{1,\mathrm{tgt}} = 0.1\, m \begin{cases} \dfrac{n^1 + N_s}{N_s - N_1} & n^1 \le -N_1 \\ -\dfrac{n^1}{N_1} & |n^1| < N_1 \\ \dfrac{n^1 - N_s}{N_s - N_1} & n^1 \ge N_1 \end{cases}, \qquad A_{2,\mathrm{tgt}} \text{ likewise in } n^2 \text{ with } N_2. \tag{4.6}$$

Theorem 4.8 (Linear response reproduces the reported depression). Suppose the output layer field has fixed shape and a time-dependent amplitude, $\mathbf{A}^{(N)}(x^0, \mathbf{x}) = a(x^0)\, \mathbf{A}_{\mathrm{tgt}}(\mathbf{x})$, and write $\rho^{(N)} = \bar{n}(1 + \delta)$. To first order in δ and $\mathbf{A}$,

$$\delta(x^0, \mathbf{x}) = \frac{e}{m}\, \nabla \cdot \mathbf{A}_{\mathrm{tgt}}(\mathbf{x}) \int_0^{x^0} a. \tag{4.7}$$

For (4.6) on the central rectangle $|n^1| < N_1$, $|n^2| < N_2$, $\nabla \cdot \mathbf{A}_{\mathrm{tgt}} = -0.1\, m^2 \left(\frac{1}{N_1} + \frac{1}{N_2} \right)$. With $a(x^0) = e^{-\gamma_2 x^0}$ and the source's $e = 0.3$, $\gamma_2 = 0.01\, m$:

$$\delta_{\mathrm{centre}}(x^0) = -0.25 \left(1 - e^{-0.01\, m x^0} \right). \tag{4.8}$$

$m x^0$	**5**	**20**	**40**	**100**	**200**
(4.8)	-1.2%	-4.5%	-8.2%	-15.8%	-21.6%
source, Figure 6	-1.3%	-4.5%	-9.1%	-16%	-21%

Proof. Linearize the third equation of (4.2): $\partial_0 \delta = \frac{e}{m} \nabla \cdot \mathbf{A}^{(N)} = \frac{e}{m} a(x^0) \nabla \cdot \mathbf{A}_{\mathrm{tgt}}$; integrate. On the central rectangle $\partial_1 A_{1,\mathrm{tgt}} = -0.1 m^2 / N_1$, $\partial_2 A_{2,\mathrm{tgt}} = -0.1 m^2 / N_2$. Then $\frac{e}{m} \cdot (-0.1 m^2)(\frac{1}{30} + \frac{1}{20}) \cdot \frac{1 - e^{-\gamma_2 x^0}}{\gamma_2} = -\frac{0.3 \cdot 0.1}{12} \cdot 100 \, (1 - e^{-0.01 m x^0})$. ■

Corollary 4.9 (What the depression is). By (4.7) the recorded quantity is the time integral of the divergence of the drive: the fluid moves with velocity $-e\mathbf{A}/m$ (Lemma 4.2) and the density follows. It is first order in e and vanishes with the coupling. It is not binary: it is the continuous number (4.8). It does not stop; it slows because the drive decays, and the drive is still at $e^{-1} = 0.37$ of its initial amplitude at $m x^0 = 100$. Nothing in (4.2) restores or holds the pattern; the source's own remark that the velocity "is nonzero at later times" and that "charged bosons will collect into the peripheral regions" is the nonlinear continuation of (4.7).

A rerun of the simulation with the corrected input amplitude 2.13 (reported in the companion checks of this folder) gives at $m x^0 = 20$ a central dip of 3.5% at $e = 0.3$, 2.1% at $e = 0.15$ and 1.1% at $e = 0.075$: halving e divides the dip by 1.7 and then 1.9, as first order in e predicts once the relative size δ of the nonlinear terms is allowed for; a second-order dependence would divide it by 4.

Theorem 4.10 (Conservation on the box). For (4.2) on a periodic box, $\int \rho^{(n)}$ is constant, and for (4.7), $\int \delta = 0$: the central depression is balanced by a rise outside. For (4.6) the rise in the corner region $|n^1| > N_1, |n^2| > N_2$ is $+0.156\,(1 - e^{-0.01 m x^0})$, i.e. $+13.5\%$ at $m x^0 = 200$.

Proof. The third equation of (4.2) is a divergence; so is (4.7). On the corners $\nabla \cdot \mathbf{A}_{\mathrm{tgt}} = 0.1 m^2 (\frac{1}{N_s - N_1} + \frac{1}{N_s - N_2})$. ■

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5. What the model establishes

Theorem 5.1. Under the hypotheses named, the model (1.1) yields exactly the following.

i. *(Pulse.)* Under H2.1–H2.7 with H2.2 replaced by Theorem 1.5 and (1.4), the reduced model (2.1)+(2.5) emits one pulse $\mathcal{A} = \frac{g \Omega N'}{4\pi} \operatorname{sech} \frac{x^0 - \tau_0}{\tau_R}$, $\tau_R = 2\pi / (g^2 \Omega N')$, provided $L_3 \ll \tau_R$; without the loss term it exchanges energy periodically with period (2.3); without the seed θ_0 nothing happens. The parameters $g, \Omega, N', L_3, \theta_0, T_2$ are not specified.

ii. *(Readout.)* Under H3.1–H3.4, a static neutral medium whose density is $c|r+o|^2$ transmits a reference wave r with the object wave present at amplitude at most $s_r = \sigma \sup|r|^2 \approx \omega_p^2/M^2$ times the object's (0.045 at the source's values), together with the twin image. The density pattern is assumed; no write exists.

iii. *(Count.)* A $2\,\mathrm{mm}$ slab of index 1.3 and area $2500\,\mathrm{cm}^2$ has 2.08×10^{16} cells of size $(\lambda/2)^2(\lambda/n_r)$ at $\lambda = 500\,\mathrm{nm}$; its surface has 4×10^{12}; the ratio is the thickness in units of λ/n_r. Cells are not bits until a write and a read are given.

iv. *(Simulation.)* Under H4.1–H4.4, a classical charged fluid with one quantum-pressure term, driven through four identical layers by an input computed by the Chebyshev recursion (4.5) (amplitude 2.13, not 3.9), develops the density change (4.7), whose central value (4.8) matches the reported Figure 6 to within 1% of density at every sampled time. The layer system does not preserve Gauss's law (4.3).

Proof. Sections 1–4. ■

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6. What the model does not contain

Let Σ be the state space of the model: triples $(\varphi, A^0, \mathbf{A})$ on the box (Section 1), or $(\rho^{(n)}, \mathbf{A}^{(n)}, \mathbf{E}^{(n)})_{n=1}^{N}$ (Section 4), with the flow Φ_t defined by the field equations and the supplied drive.

Definition 6.1 (Memory). A memory over an object set $\mathcal{O}$ consists of a write $W : \mathcal{O}\times\Sigma \to \Sigma$, a hold time τ_{ret} with $\|\Phi_t W(o,\sigma) - W(o,\sigma)\| \le \epsilon$ in a norm on Σ for $0 \le t \le \tau_{\mathrm{ret}}$, and a read $R : \Sigma \to \mathcal{O}$ with $R(\Phi_t W(o,\sigma)) = o$ for all $t \le \tau_{\mathrm{ret}}$ except on a set of error rate p.

Proposition 6.2. In Sections 1–4: the only input to Σ is the supplied drive $\mathbf{A}^{(0)}$ (Section 4) or the supplied pattern (H3.4), neither of which is a map from an object set; no read R is defined; no hold time follows from the equations (Corollary 4.9, Lemma 3.9); and no map from Σ to any set other than Σ itself appears. Hence no statement of the form "the pattern is recalled", "holograms correspond one-to-one to X", or "X is manipulated", for any X outside Σ, is a proposition about the model: it has no referent in Σ. ■

Open problems. Each is a theorem that would have to be proved, or a number that would have to be supplied, before Theorem 5.1 says anything about a medium.

- **O1.** Values of $g, \Omega, N', L_3, \theta_0, T_2$ for a stated medium, and a proof of (2.9).
- **O2.** A bound on the Coulomb term (1.4) for that medium, or a proof that the background ρ_b neutralizes it.
- **O3.** A write: an equation whose solution has $\rho \propto |r+o|^2$ with o an input, from (1.1) or a stated extension.
- **O4.** A read: a map R on Σ with a proved error rate.
- **O5.** A hold: a barrier ΔE for the pattern, from the equations, with (3.8) evaluated.
- **O6.** The ablation: the solution of (4.2) with the $\hbar^2$ term removed, compared with Figures 5–6; and the solution with (G) imposed, compared with (4.3).

Appendix A. Hypothesis ledger

Label	Statement	Where used	Status
ρ_b	background charge density	§1, §3	supplied; 0 in the source; required $\neq 0$ on a torus (Thm 1.5)
H2.1	two real levels, $\Omega = E_1 - E_0$	§2	supplied
H2.2	$A^0 = 0$, $\mathbf{A} \parallel \hat{\mathbf{e}}_3$	§2	supplied; contradicts a changing density (Thm 1.4, Thm 2.9)
H2.3	resonant shell, $\Omega^2 \geq \omega_p^2$	§2	supplied; threshold corrected from the source
H2.4	slow envelopes, $\varepsilon < \frac{1}{2}$	Prop 2.8	supplied (rotating-wave approximation)
H2.5	dipole, $L_{1,2} \ll 2\pi/\sqrt{\Omega^2 - \omega_p^2}$	§2	supplied; fails for any emitting region larger than the wavelength, such as the 2500 cm^2 slab of §3 at 500 nm
H2.6	loss $-\mathcal{A}/L_3$	Thm 2.6	supplied model
H2.7	seed $\theta_0 \neq 0$	Cor 2.7	supplied; undefined in the source
(1.4)	$e \sup\lvert A^0 \rvert \ll \Omega$	Thm 2.9	derived condition; not evaluated by the source
(2.6)	$L_3 \ll \tau_R$	Thm 2.6	derived condition; not evaluated
H3.1	static neutral medium	§3	supplied; the one consistent use of H2.2
H3.2	screening M^2, source $M^2\tilde{\mathbf{A}}$	§3	supplied constants
H3.3	polarization $\hat{\mathbf{e}}_3$	§3	supplied
H3.4	$\rho = c\lvert r + o \rvert^2$	Thm 3.4	supplied; no write (Rem 3.6); o has no source
H3.5	one cell = one bit	Prop 3.8	supplied; needs a write and a read
H4.1	planar, $U = 0$	§4	supplied
H4.2	M^2, γ replace the self-energy	§4	supplied constants (2 and 0.2 in units of m)
H4.3	four identical layers, coupling v, drive $\mathbf{A}^{(0)}$	§4	supplied; drive computed from the target
H4.4	(G) not imposed	Thm 4.5	consequence of the unitary gauge; breaks Gauss's law
(4.6)	targets	§4	supplied by the authors
m	the mass scale	§4	never given; at 4 meV the lattice cell is 49 μm, at the mass of a water molecule 10^{-17} m

Appendix B. Corrections to the source

Location	As printed	Corrected
before (A2)	"Coulomb gauge, namely $A^0 = 0$ and $\nabla_i A_i = 0$"	two conditions, not a gauge; freezes ρ (Thm 1.4); Coulomb gauge is Thm 1.5
(A31)	χ_1 on the right	χ_0
after (A29), after (A47)	$\Omega \geq e^2\lvert\bar{\varphi}\rvert^2/m$	$\Omega^2 \geq e^2\bar{n}/m$
§2	"definition of θ_0 is in Appendix A"	θ_0 is a hypothesis (H2.7)
(A52)	$\tilde{A}_i \propto \cdots$	$A_i - \tilde{A}_i = -\sigma[\cdots]$, (3.5)
§3	$k_B T \ln 2$ bounds thermal degradation	it bounds erasure heat; retention needs (3.8)
(A70)	$-\frac{e}{2m}(A_i - \partial_i\beta/e)^2$	$+$, as in (A81), (A87) and (B)
(A76), (A78)	$e\lvert\bar{\varphi}\rvert^2/m$	$e^2\lvert\bar{\varphi}\rvert^2/m$
(A88), (A89)	$-\partial_1^2 u_1$, $-\partial_2^2 u_2$	$-\partial_2^2 u_1$, $-\partial_1^2 u_2$, (4.4)
§4	input maximum $0.1 \times 2^4/0.8^4 = 3.9$	$0.1\,U_4(1.25) = 2.13$ (Cor 4.7)
§4	"distributions … stop evolving"	(4.8): they stop because the drive decays
§5	"one-to-one correspondence"	no second set exists (Prop 6.2)

Appendix C. Numbers

Quantity	Value	From
$k_B T$ at 310 K	26.71 meV	Prop 3.10
$hc/(k_B T \ln 2)$	67 μm	Prop 3.10
$k_B T \ln 2$	18.52 meV	Prop 3.10
barrier for 1 s, 1 h, 1 yr ($\tau_a = 1$ ps)	$0.74, 0.96, 1.20$ eV	(3.8)
C_2, $n_r d/\lambda$, C_3 at 500 nm	$4.0 \times 10^{12}, 5200, 2.08 \times 10^{16}$	Prop 3.8
C_3 at 1500 nm, at 310 μm	$7.7 \times 10^{14}, 8.7 \times 10^{7}$	Prop 3.8
$s = \omega_p^2/M^2$ at the Section 4 values	0.045	Cor 3.5
$U_4(1.25), 0.1\,U_4$	$21.3125, 2.13$	Cor 4.7
$U_4(1.30625), 0.1\,U_4$	$27.11, 2.71$	Cor 4.7
δ_{centre} at $mx^0 = 5, 20, 40, 100, 200$	$-1.2, -4.5, -8.2, -15.8, -21.6$ %	(4.8)
δ_{corner} at $mx^0 = 200$	$+13.5$ %	Thm 4.10
$T(\theta_0)$ for $\theta_0 = 0.1$	$4.38 \times (4/\omega_c)$	(2.3)

Appendix D. Verification map

Every result below is recomputed from scratch by the companion script `7_paper_rewritten_checks.py` (numpy and sympy only; it prints PASS or FAIL per line and exits with status 1 on any failure). Symbolic checks prove an identity for arbitrary functions; numerical checks integrate the stated equations and compare with the stated closed forms.

Result	Check line	Method
Prop 1.2	Prop 1.2	symbolic: kinetic-term expansion and $\partial\mathcal{L}/\partial A_i = e\rho\mathcal{V}_i$
Thm 1.3	Thm 1.3	symbolic: $\nabla\cdot(\nabla^2\mathbf{A} - \nabla\nabla\cdot\mathbf{A}) = 0$
Prop 1.6	Prop 1.6	numerical: slab comparison bound; ball integral $2\pi a^2$
Prop 2.3	Prop 2.3	numerical: invariants along (2.1) to 10^{-13}
Thm 2.4, 2.5 (i)–(ii)	Thm 2.4/2.5	numerical: pendulum equation, period (2.3), peak, turning points
Thm 2.5 (iii)	Thm 2.5(iii)	numerical: sech limit (2.4), logarithmic period
Thm 2.6	Thm 2.6	numerical: (2.5) at $L_3/\tau_R = 0.005$ against (2.8)
Cor 2.7	Cor 2.7	numerical: fixed point
Prop 2.8	Prop 2.8	symbolic: density expansion
Thm 3.2	Thm 3.2	symbolic: residual (3.3)
Thm 3.4, Cor 3.5	Thm 3.4	symbolic: (3.5); $s = 0.045$
Prop 3.8	Prop 3.8	arithmetic
Prop 3.10	Prop 3.10	arithmetic
Thm 4.1	Thm 4.1	symbolic: (C) and (B); printed (A70) sign fails on a plane wave
Thm 4.5	Thm 4.5	symbolic: (4.3)
Thm 4.6, Cor 4.7	Thm 4.6/Cor 4.7, Thm 4.6 (lattice)	recursion, closed form, forward solve; derivative terms on the lattice
Thm 4.8	Thm 4.8	lattice divergence of (4.6), table against Figure 6
Thm 4.10	Thm 4.10	lattice sum, corner rise
Appendix A, C	App. A/C	arithmetic

Not replayed by the script: the Gauss-residual ratio quoted after Theorem 4.5 and the three rerun dips quoted after Corollary 4.9, which come from reruns of the source's full simulation reported elsewhere in this folder.